

Student 30 **Suba Shree K** 192521396RD

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Q1. LIST**ANSWER**

List comprehension offer a shorter syntax. It helps in reducing lines of code when you want to mathematically generate a new list based on the values of an existing iteration .stores all elements in a list, so it uses more memory.

Example :

```
a=[x*x for x in range(1,6)]
```

```
print(a)
```

OUTPUT:

```
[1,4,9,16,25]
```

Generator expression:

It produces elements once at a time when it is needed so it uses less memory.

example:

```
b=[x*x for x in range(1,6)]
```

```
print(next(b))
```

```
print(next(b))
```

output:

```
[1,4]
```

Q2. LIST**ANSWER**

```
num=[1,2,3,4,5,6]
```

```
result=[0 if x%2==0 else x for x in num]
```

```
print(result)
```

OUTPUT:

```
[1,0,3,0,5,0]
```

Q3. LIST**ANSWER**

```
n=[1,2,3,2,4,2]
```

```
result=[x for x if n!=2]
```

```
print(result)
```

OUTPUT:

```
[1,3,4]
```

Q4. Tuple**ANSWER**

=>A Tuple is essentially a list that can have values inside () bracket. Once the values are created we cannot add remove or change its items.

=>No, tuple cannot contain mutable elements.

=>Example:

```
num=(1,2,3,4,5,6,7,8,9)
```

```
print(num)
```

OUTPUT:

```
(1,2,3,4,5,6,7,8,9)
```

Q5. SET**ANSWER**

A frozen set is a set where set is cannot be changed or modified. Here the frozen set cannot even append the values or delete the values. It have values inside the curly braces {}, unlike a dictionary. Frozen set is immutable ,because the values are assigned in a fixed position. And we cannot replace it.

Example:

```
n=frozenset([1,2,3,4,5])
```

```
print(n)
```

OUTPUT:

```
((1,2,3,4,5,5))
```

```
//since frozen set cannot be changed
```